

# Functions\_EXP

`__init__(model_dir=None, data_dir=None)`

- initialize needed stuff

`init_csv_files()`

- **Role:** Ensures `questions.csv` and `answers.csv` exist. If `questions.csv` is missing, it populates it from the raw question bank and calculates initial text complexity features.

`get_question_pool()`

- Loads the available questions and encodes their `difficulty` and `topic` into numbers for the model.

`score_pool(student_id, history, pool)`

- calculates the probas

`select_next_question(strategy, probas, pool)`

- **Returns** the index of the selected question depending on the strategy

`should_stop(probas)`

- checks for the thresholds to stop when the model is certain

`create_student_instance()`

- Initializes a new instance of student with default values

`record_answer(student_id, question_id, student_answer, response_time)`

- Checks for student answer and updates the features

`prepare_the_query(student_id)`

- **Role:** Similar to `score_pool`, but reads the student's history directly from `answers.csv`. (might or might not be needed)

## internal helper methods

---

- `_get_metrics_from_history` : Calculates averages (topic correctness, avg response time) from a list of interactions.
- `_build_vector_internal` : constructs the feature vector used by the model.
- `compute_question_features` : Uses `textstat` to calculate the complexity grade of a question's text.